

Homework for week 2, submission deadline 23:59 on Sunday night.

1. Choose your own topics from following list or your own.

- Gut microbiota and metabolic disease
- Single-cell transcriptomics in cancer immunology
- Antibiotic resistance genomics
- Epigenetic regulation in neurological disorders
- Muscle regeneration and aging
- Proteomic biomarkers in MAFLD/MASH

Finish the following question section we covered in the class

#	Student Action	Output
1	Write a broad topic	One phrase
2	Narrow to a phenomenon	One sentence
3	State a testable hypothesis	One falsifiable claim
4	Apply PICO/PECO	Four structured elements
5	Identify a dataset or data type	One database or accession
6	Draft a minimal analysis plan	Data, method, output, validation

1、topic: 螺旋藻用于高价值化合物生产的代谢工程

2、phenomenon: 在光自养培养条件下, 过表达关键生物合成基因可提高钝顶螺旋藻的藻蓝蛋白产量。

3、hypothesis: 在钝顶螺旋藻中过表达 *cpcA-cpcB* 操纵子能够显著提升细胞内藻蓝蛋白含量; 若改造株与野生株藻蓝蛋白含量无统计学显著差异, 则该假设不成立。

4、PECO: Population: 钝顶螺旋藻野生型菌株

Exposure: 基因改造: 过表达藻蓝蛋白合成操纵子 *cpcA-cpcB*

Comparator: 未做转化的钝顶螺旋藻野生对照株

Outcome: 光自养培养下胞内藻蓝蛋白含量、生物量生长速率

5、dataset: NCBI GenBank: Genome accession: GCA_000224545.1 (*Arthrospira platensis* NIES-39 genome); 实验数据类型: 转录组 RNA-seq、紫外-可见分光光度蛋白定量、生长曲线 OD 监测数据

6、analysis plan:

①Data: Reference genome GCA_000224545.1; RNA-seq 转录组测序读数; 分光光度法藻蓝蛋白测定值; OD 生长曲线数据。

②Method: 构建 *cpcA-cpcB* 过表达质粒, 转化野生钝顶螺旋藻; 改造株与对照组在相同光自养条件培养; 转录组测序; 定量藻蓝蛋白与生物量; DESeq2 差异基因分析, t 检验表型统计。

③Output: 差异基因表达谱; 藻蓝蛋白定量结果; 改造株与野生株生长曲线对比。

④Validation: 设置生物学三次重复; qPCR 验证 *cpcA/cpcB* 转录水平; 独立批次重复培养验证藻蓝蛋白表型。

2. Find and read two research papers in the folder and answer following question based on these two papers.

- Question 1: Use the PICO/PECO, what are they for these two papers?
- Question 2: What kind of dataset or data type these two papers use?
- Question 3: What is the hypothesis of these two research?

Paper1:

a.PECO:①Population: 2021 年 6–10 月在珠江医院招募的 34 对母婴 (MIPs): 34 名足月顺产、纯母乳 / 以母乳为主、出生后 100 天内未使用抗生素的婴儿, 收集出生 (胎便) 及出生后第 1、2、3 月共 4 个时间点粪便样本。

②Exposure: 早期肠道微生物定植与菌群组成 (16S rRNA 基因 V3–V4 区扩增子测序); 关注菌群网络结构、 α 多样性 (InvSimpson)、 β 多样性 (PCoA)、分子生态网络 (MENs) 拓扑特征; 同时考察母亲孕晚期 TG、BMI 等。

③Comparator: 对照组: 6 月龄内未发生湿疹的健康婴儿 (n=22); 验证队列: Roswall 等 (瑞典 16 名)、Christensen 等 (丹麦 16 名) 健康婴儿。

④Outcome: 至 6 月龄是否发生婴儿湿疹; 以及菌群网络拓扑指数 (簇数、边数、正 / 负边、顶点数、平均路径长度等) 的动态变化是否出现异常 ("节律丢失")。

b.

Datatype: 16S rRNA 基因扩增子测序数据 (V3–V4 区, Illumina NovaSeq 平台, 平均每样本 88,405 条读段; 共 140 份粪便样本, 136 份用于分析)

Dataset: 国家微生物科学数据中心 NMDC

Accession: NMDC10018172

c. hypothesis:

①健康婴儿的早期菌群网络随月龄呈现有节律的复杂性上升；而湿疹婴儿网络出现 "节律丢失"

②健康婴儿的肠道菌群网络比湿疹婴儿更致密、更复杂、更稳定

Paper2:

a.PECO:

①Population: 三种胃肠道疾病患者的结肠黏膜活检样本，共 208 对配对的宿主转录组 + 微生物组样本 (416 样本): CRC 88 对 (44 肿瘤 + 44 配对正常)、IBD 78 对 (56 患者 + 22 对照)、IBS 42 对 (29 患者 + 13 对照)。

②Exposure: 不同疾病状态 (CRC / IBD / IBS) 及对应的肠道微生物组组成 (16S rRNA 测序的菌群丰度); 分析宿主基因表达与微生物组丰度之间的关联。

③Comparator: 各队列内的非疾病对照: CRC 配对正常结肠组织、非 IBD (n=22)、非 IBS (n=13) 对照组。

④Outcome: 宿主基因-微生物组关联: 共享与疾病特异性的宿主通路 (如 RAC1、氧化磷酸化、整合素通路)、具体宿主基因-菌群互作网络 (如 Streptococcus、Peptostreptococcaceae、Blautia)。

b.

Datatype: 宿主转录组 RNA-seq (结肠黏膜活检) + 肠道微生物组 16S rRNA 测序 (菌群丰度)

Dataset: NCBI SRA 数据库

Accession: BioProject Accession: PRJNA816986

c. hypothesis:

①在 CRC、IBD、IBS 中，宿主基因表达与肠道微生物组成之间存在共享与疾病特异性的关联;

②相同的微生物在不同疾病中可通过调控不同的宿主基因，以疾病特异性的方式影响宿主病理生理 (如 Streptococcus 在 CRC 关联 WNT/NF- κ B 通路基因、在 IBS 关联巨噬细胞炎症基因)。

3. Based on your research interest, refine the search in PUBMED, and sort, identify the latest and the most important studies in your field. Read this paper and answer three questions:
- Question 1: Use the PICO/PECO, what are they for these two papers?
 - Question 2: What kind of dataset or data type these two papers use?
 - Question 3: What is the hypothesis of these two research?

选择文献:

Title: Cultivation of *Arthrospira platensis* in seawater culture media for CO₂ capture and production of valuable compounds

Journal: Journal of Environmental Management

Year: 2025

DOI: 10.1016/j.jenvman.2025.127845

Article type: Original experimental research article

a.PECO:

- ①Population: *Arthrospira platensis* strain SAG 21.99
- ②Exposure: Two seawater-based cultivation interventions carried out in laboratory-scale photobioreactors with continuous ambient CO₂ supply for total 28-day batch culture.
 1. ZSW group: Seawater-based modified Zarrouk medium with full Zarrouk nutrient formulation prepared with filtered natural seawater, using salinity-acclimatized SZI inoculum.
 2. NSW group: Filtered natural seawater supplemented only with nitrate, phosphate and trace-element nutrients, without full Zarrouk chemical system.
- ③Comparator: ZCE control group, conventional freshwater-based complete Zarrouk medium, using standard ZI inoculum.
- ④Outcome: 1、CO₂ capture / bio-fixation efficiency and daily CO₂ fixation rate;
2、Biomass productivity, final biomass concentration and biofilm formation phenotype;

3、 Intracellular biochemical composition: protein, carbohydrate, total lipid content;

4、 Photosynthetic accessory pigments: C-phyococyanin, C-allophycocyanin and chlorophyll-a concentration.

b.

Datatype: Phenotypic and biochemical experimental data, including biomass gravimetry, real-time CO₂ monitoring, colorimetric protein/carbohydrate/lipid assays, spectrophotometric pigment quantification, and statistical analysis results.

Dataset: In-house lab phenotypic dataset(无测序、多组学原始数据); raw data can be requested from the corresponding author.

Accession: NCBI GenBank: GCA_000224545.1

c.hypothesis: Seawater-based media with appropriate nutrient supplementation can support salinity-acclimatized *Arthrospira platensis* to achieve comparable CO₂-fixation capacity and competitive yields of high-value bioproducts, relative to the conventional freshwater Zarrouk medium.